

Reusable architecture knowledge

Prior projects + literature

design artifacts + documents
tool traces + measurements
failures + reviewed results

Current project record

Versioned study state

workload + software versions
constraints + candidate identity
tool, library + process versions
current measurements + failures

Architecture role

supplies intent + constraints
reviews result and forms a
supported technical recommendation

Commitment authority

decides advance, hold, or reject

AI-assisted architecture design system



Architecture foundation model

optional learned component
adapted to the current task



Conventional methods

models, algorithms, solvers



Design tools

compilers + simulators
RTL, EDA + formal tools

Architecture tasks

workload + requirements analysis

prediction + design-space screening

candidate, software + RTL generation

diagnosis + verification assistance



Measurements and checks

qualified tool returns + failures
assumptions + validity limits
independent rejection conditions



Supported architecture result

comparison + measurements + assumptions + checks + limits

The moonshot is complete-system capability, not autonomous prompt-to-chip generation.